

TEZPUR UNIVERSITY
Mid Semester Test, Autumn 2018
C0 504 Natural Language Processing

Marks : 40

Time : 75 minutes

The figures in the right-hand margin indicate marks for the individual question.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

1. Write a Perl code segment that selects lines that start with the word “What”? 2
2. (a) How are the following distinct- Unicode, UTF-16, and “Times New Roman” fonts? 4
 (b) What is *sampling* of a signal? 3
3. Why is *smoothing* done? What is *add-one smoothing*? 2 + 2
4. Consider the following text-

The reporters know that he can talk very fast.
 They have come ready with their questions
 and recording devices.

 - (a) Suggest a Parts-of-Speech tagset sufficient to depict the challenge of PoS tagging task with respect to the above text? 3
 - (b) Depict the challenge of PoS tagging task with respect to the above text? 2

- (c) Perform PoS tagging of the above text manually using the tagset you have suggested. 3
 - (d) Suggest a rule that will help tag the word “fast” correctly. 2
5. Give an example of *left recursion* in a context free grammar? 1
 6. (a) Give examples of noun phrases and prepositional phrases. 3
 (b) Give examples of a declarative and an imperative sentence in English and suggest a context free grammar for such sentences. 2 + 2
 7. Briefly explain the phenomenon of *agreement* in English sentences. For syntax analysis why is this phenomenon not handled by provisions of Context Free Grammar? 2 + 2
 8. Consider the following sentence-

I saw him waiting for the bus.

 - (a) Show how the above sentence is ambiguous. What kind of ambiguity is there? 3 + 1
 - (b) What would be the output of Earley’s parser for an ambiguous sentence like the one above? 1